AUTOSAR HIGH-LEVEL DESIGN (HLD) SPECIFICATION

Document ID: HLD-SWC-VSM-002 | Version: 1.2 | Status: DRAFT FOR REVIEW
Component: VehicleSpeedManagement_SWC (Application SWC)
ASIL Rating: ASIL-B | Target Platform: AUTOSAR Classic Platform 4.4

1. ARCHITECTURAL OVERVIEW

The VehicleSpeedManagement_SWC is responsible for calculating filtered vehicle speed from wheel pulse sensors and broadcasting the validated speed to the Powertrain and Dashboard clusters via the RTE.

Software Component Type: Application SWC
Connected SWCs: WheelSpeedSensor_SWC, InstrumentCluster_SWC, EngineControl_SWC
BSW Dependencies: ComStack (CanIf, CanTp, Com), Dem (Diagnostic Event Manager), NvM

2. PORT INTERFACES & DATA ELEMENTS

The component defines the following AUTOSAR Ports:

1. P-Port: ppVehicleSpeed
   - Interface Type: Sender-Receiver Interface (IF_VehicleSpeed)
   - Data Element: VehicleSpeed_kmh (uint16, 0.01 km/h resolution)
   - Transmission Mode: Periodic, queued=False

2. R-Port: rpWheelPulseRaw
   - Interface Type: Sender-Receiver Interface (IF_WheelPulseData)
   - Data Elements: LeftFrontPulse, RightFrontPulse, LeftRearPulse, RightRearPulse

3. R-Port: rpDiagStatus
   - Interface Type: Client-Server Interface (IF_DiagStatusService)
   - Operation: GetFaultStatus(out FaultMask: uint8)

3. RUNNABLES & RTE EVENT MAPPINGS

Runnable Entities defined within the SWC:

1. Runnable: RE_CalculateSpeed
   - Trigger: RTE Timing Event (Period: 10ms)
   - OS Task: Task_Periodic_10ms (Preemptive, Priority 12)
   - Description: Reads rpWheelPulseRaw, executes filtering algorithm, writes to ppVehicleSpeed.

2. Runnable: RE_VSM_Init
   - Trigger: RTE Init Event (swc_init)
   - Description: Initializes internal calibration parameters and fault masks.

3. Runnable: RE_DiagEventUpdate
   - Trigger: Data Received Event on rpWheelPulseRaw
   - Description: Detects signal plausibility failures and reports DTC to Dem.

4. BSW & SAFETY REQUIREMENTS

Functional Safety & Diagnostics:
- Safety Standard: ISO 26262 ASIL-B
- E2E Protection: Profile 2 (CRC-8 + Sequence Counter) applied to IF_VehicleSpeed.
- Diagnostic Monitoring: Dem_SetEventStatus called if wheel sensor differential > 25 km/h.
- Memory Mapping: SwAddrMethod VAR_FAST for speed calculation buffer variables.

Architectural Verification Checklist:
[x] All R-Ports mapped to valid provider P-Ports via Composition connectors.
[x] Runnable execution periods aligned with OS Task activation periods.
[ ] Server response timeout for rpDiagStatus Client-Server invocation (Under Review).